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B.TECH CSE PROJECT DOCUMENTATION REPORT

FuelScope – Global Fuel Intelligence

Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report (2020–2026)

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Executive Summary

FuelScope – Global Fuel Intelligence is an end-to-end data analytics and visualization platform designed to study international fuel pricing structures, taxation policies, and economic indicators from 2020 to 2026. Encompassing 27,468 records across 84 countries and 7 global regions, this project provides multi-perspective analytical stories and interactive web interfaces to evaluate petrol, diesel, and LPG trends for policy makers, supply chain leaders, and energy researchers.

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Chapter 1: Introduction & Problem Statement

1.1 Project Introduction

Fuel pricing dynamics represent a critical pillar of macroeconomics, directly impacting transportation costs, consumer price indices, industrial manufacturing, and overall national economic stability. **FuelScope – Global Fuel Intelligence** is developed as a comprehensive analytical framework to evaluate how global fuel prices—specifically petrol, diesel, and Liquefied Petroleum Gas (LPG)—fluctuate under the influence of government taxation policies, national subsidies, and macro-level income tiers across the globe.

1.2 Problem Statement

Global energy markets are inherently complex and fragmented. Variations in national tax structures, environmental policies, geopolitical tensions, and local subsidy regimes create substantial discrepancies in retail fuel prices across geographical regions. Raw energy metrics are often distributed across multiple government feeds and international repositories without unified standardization.

Without an interactive, consolidated platform, policy analysts, logistics directors, and researchers struggle to synthesize cross-country pricing mechanisms. There is a distinct necessity for an intuitive data intelligence tool that aggregates historical multi-fuel data, contextualizes tax-versus-subsidy tradeoffs, and presents persona-driven analytical stories.

Chapter 2: Project Objectives & Scope

2.1 Core Project Objectives

- **Analyze Global Pricing Variations:** Evaluate price dispersion of petrol, diesel, and LPG across 84 countries and 7 geographic regions.
- **Cross-Fuel Comparative Analysis:** Conduct comparative studies between commercial fuel types to identify pricing disparities and regional preferences.
- **Taxation Structure Examination:** Detail differences in fuel taxation levels across sovereign nations to identify high-tax vs low-tax economic regimes.
- **Subsidy Impact Assessment:** Study the structural relationships between state-level fuel subsidies and retail pricing dynamics.
- **Longitudinal Trend Tracking:** Track longitudinal fuel price transformations spanning the six-year observation window from 2020 to 2026.
- **Multi-Persona Storytelling:** Deliver specialized interactive dashboards and tailored analytical stories catering to distinct professional domain perspectives.

Chapter 3: Dataset Description & Preprocessing

3.1 Dataset Overview

The underlying dataset of the FuelScope project offers a robust repository of multi-national fuel economy indicators. The data provides longitudinal granularity spanning six years, allowing for robust statistical evaluation.

Dataset Attribute	Specification Details
Temporal Scope	2020 – 2026 (6-Year Longitudinal Window)
Geographic Coverage	84 Sovereign Countries
Regional Classification	7 Global Regions
Total Volume	27,468 Complete Observations
Commodity Coverage	Petrol, Diesel, Liquefied Petroleum Gas (LPG)
Socio-Economic Factors	Fuel Taxation, Government Subsidies, Income Class, Economic Indicators

3.2 Data Preprocessing & Cleaning

Prior to visual deployment in Tableau Public, the raw data underwent systematic preprocessing to guarantee analytical accuracy and pipeline integrity:

- **Data Validation & Integrity Verification:** Evaluated all 27,468 records for missing values, structural anomalies, and duplicate entries.
- **Currency & Measurement Standardization:** Standardized multi-currency pricing formats into unified international standard units to enable cross-country comparison.
- **Taxation & Subsidy Mapping:** Categorized raw tax figures and subsidy amounts into distinct analytical brackets for comparative evaluation.
- **Categorical Hierarchy Establishment:** Defined parent-child relationships linking individual countries to their respective 7 overarching global regions.

Chapter 4: Visualizations & Tableau Links

The core intelligence of FuelScope is powered by Tableau Public interactive visual engines. The project features one primary global dashboard and three specialized analytical stories designed for key industry personas.

Main Dashboard: Global Fuel Prices and Taxation Trends (2020–2026)

Consolidated interactive overview tracking global fuel price indices, comparative tax burden maps, and regional trends.

Access Main Tableau Dashboard

Story 1: Dr. Sarah Mitchell – Energy Policy Consultant Perspective

Focuses on regulatory policy frameworks, national fuel taxation variations, carbon tax impacts, and sovereign subsidy structures.

Access Policy Consultant Story

Story 2: Marcus Thompson – Global Supply Chain Manager Perspective

Centers on operational fuel procurement, regional logistics cost variances, diesel vs. petrol trade-offs, and supply chain vulnerability.

Access Supply Chain Story

Story 3: Professor Elena Rodriguez – Energy Economics Researcher Perspective

Investigates macro-economic correlates, income level distributions, multi-year inflation effects, and econometric fuel models.

Access Energy Economics Story

Chapter 5: Key Features & Technical Architecture

5.1 Core Platform Features

- **Interactive Multi-Fuel Dashboards:** Embedded responsive Tableau dashboards for dynamic user filtration.
- **Regional & Sovereign Filtering:** Ability to isolate data by continent, regional block, or individual country.
- **Tax & Subsidy Spectrum Analysis:** Analytical identification of top tax-imposing economies versus heavily subsidized fuel markets.
- **Commodity Comparison Engine:** Direct comparative visualization of Petrol vs. Diesel vs. LPG pricing curves.
- **Persona-Based Storytelling:** Pre-built analytical journeys curated for policy makers, logisticians, and researchers.

5.2 Technical Stack Overview

Technology Component	Role & Purpose in Project Architecture
HTML5	Provides semantic structure and document hierarchy for the core web web application.
CSS3	Custom responsive styling, modern layout grids, and academic color palette implementation.
JavaScript (ES6+)	Manages client-side DOM interaction, smooth dynamic behaviors, and visual embed handling.
GSAP (GreenSock)	Delivers smooth scroll animations, hero section motion effects, and sleek page transitions.
Tableau Public	Powers data transformation, analytical charting, interactive filters, and story rendering.
Git / GitHub	Version control system for tracking source code revisions and team collaboration.
GitHub Pages	Static web hosting infrastructure for public deployment of the interactive platform.

Chapter 6: Development Process & Methodology

1. **Problem Identification & Domain Research:** Formulated analytical scope surrounding global energy pricing fluctuations and policy variances.
2. **Data Acquisition & Structuring:** Curated the comprehensive 27,468-record dataset spanning 84 countries and 2020–2026 timelines.
3. **Data Cleaning & Transformation:** Cleaned nulls, standardized measures, and verified categorical tags for seamless BI integration.
4. **Exploratory Data Analysis (EDA):** Examined underlying statistical distributions, regional skewness, and tax-to-price correlations.
5. **Visualization & Dashboard Design:** Built specialized worksheets, geographical heatmaps, and consolidated interactive dashboards in Tableau.
6. **Analytical Story Construction:** Authored three targeted narratives addressing distinct operational and research queries.
7. **Frontend Web Development:** Constructed the dark-blue modern web container using HTML5, CSS3, JS, and GSAP for interactive visualization embedding.
8. **Deployment & Verification:** Hosted site assets on GitHub Pages and verified cross-browser compatibility and link integrity.

Chapter 7: Analytical Scope & Exploration Areas

The FuelScope system enables stakeholders to explore several key empirical relationships across energy markets:

- **Cross-National Price Disparities:** Exploring why geographically adjacent nations exhibit contrasting retail fuel rates.
- **Tax Heavy Economies:** Identifying nations where tax policies constitute a dominant fraction of the total pump price.
- **Commercial Fuel Divergence:** Studying differential pricing between commercial diesel and retail petrol across different regulatory environments.
- **Macroeconomic & Income Correlations:** Observing how high-income versus developing economic tiers absorb energy cost shocks over time.

Chapter 8: Local Setup & Deployment Guide

8.1 Steps for Running Locally

1. **Clone repository:** Clone or download the source repository using Git: `git clone https://github.com/[username]/FuelScope.git`
2. **Navigate to Directory:** Open the extracted project folder on your local machine.
3. **Launch Application:** Locate the main `index.html` file and double-click or open it inside any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge).
4. **Network Connectivity:** Ensure an active internet connection is available to allow external Tableau JavaScript API assets and embedded visualizations to render seamlessly.

Chapter 9: GitHub Pages Cloud Deployment

1. Create a public GitHub repository named `FuelScope`.
2. Push all project source files, including `index.html`, CSS stylesheets, JS assets, and images, to the `main` branch.
3. Navigate to repository **Settings** and click on the **Pages** tab on the left sidebar.
4. Under **Build and deployment**, select **Deploy from a branch**.
5. Set the branch to `main` and folder to `/(root)`, then click **Save**.
6. Wait a few minutes for GitHub Actions to complete the deployment, then copy the live site URL provided.

Chapter 10: Conclusion

FuelScope – Global Fuel Intelligence successfully integrates robust data analytics, interactive Tableau visualization, and modern web engineering into a unified academic and practical software project. By organizing complex fuel pricing and taxation data spanning 84 countries into clear, persona-driven analytical stories, the project demonstrates how modern front-end technologies and data visualization frameworks can make complex global energy metrics accessible, insightful, and actionable.